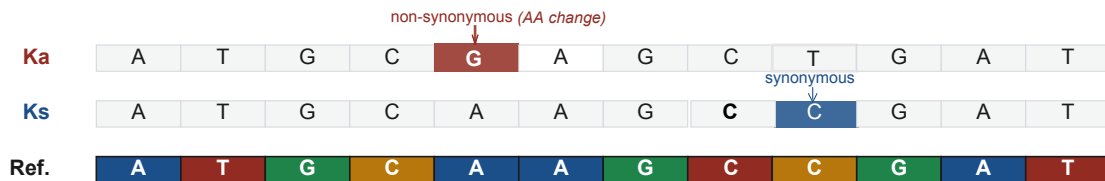


(a)



$$\omega = \frac{K_a}{K_s}$$

 $\omega > 1$: positive selection

 $\omega \approx 1$: neutral drift

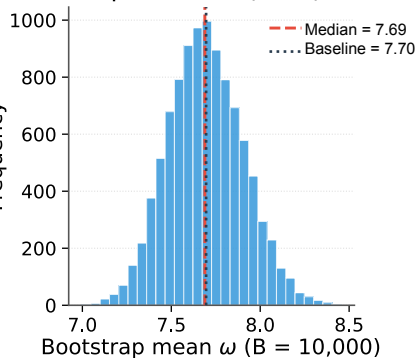
 $\omega < 1$: purifying selection

$$\text{CKI } \omega = \frac{k_f}{k_n} = \frac{\text{JS}(\text{softmax}(\epsilon_A[I]), \text{softmax}(\epsilon_B[I]))}{\text{JS}(\text{softmax}(\epsilon_A[H]), \text{softmax}(\epsilon_B[H]))} \quad \omega: \text{transcriptomic divergence relative to the baseline}$$

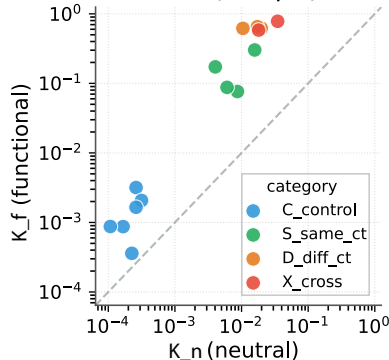
(b)



(c)

Split-half control ω (n = 300)

(d)

 K_n VS K_f (mouse pilot)

(e)

 ω distribution (n = 15)